

01 / HIGH-STAKES COMPLIANCE

Why a 2mm piping error can disqualify an Olympic rider

In high-performance equestrian wear, garments are governed as regulated athletic equipment by the Fédération Équestre Internationale (FEI).

GARMENT SPECIFICATION: Sovereign Grand Prix Coat

Discipline: Olympic Show Jumping | Material: 75% Polyamide, 25% Elastane

DISQUALIFICATION RISK

FEI Jumping Rules, Art. 256.1.4

Collar Sponsor Logo Area: 65.0 cm²

Regulatory Ceiling: <= 60.0 cm² | Delta Overage: +5.0 cm² (+8.3%)

DISQUALIFICATION RISK

FEI Dressage Rules, Art. 427.1

Collar Velvet Piping Width: 3.5 mm

Regulatory Ceiling: <= 3.0 mm | Delta Overage: +0.5 mm (+16.7%)

PASS

Fabric Breathability: 12,000 g/m²/24h (SOP threshold: >= 10,000)
Hydrostatic Head: 4,000 mm (Water resistant athletic standard)

Manual Tech Pack Audit: 2 to 4 hours -> Atelier Pro: < 0.35 seconds

Why LLMs fail at high-stakes compliance math

Probabilistic token predictors hallucinate mathematical bounds and invent fake rules. In athletic regulations, you cannot bet an Olympic career on probabilities.

GENERIC AI WRAPPER

Prompting an LLM Directly

- X Fuzzy Math**
Decides 65 cm² is 'close enough' to 60 cm² based on conversational training weights.
- X Article Hallucination**
Quotes nonexistent rules (e.g. 'FEI Art. 302.4') that sound authoritative but do not exist.
- X Non-Deterministic**
Running the exact same tech pack three times produces conflicting compliance verdicts.
- X High Latency & Cost**
Spends 15 to 30 seconds reading entire PDFs via expensive multi-shot prompt chaining.

DETERMINISTIC VERIFIER

Our Dual-Layer Architecture

- / Decoupled Perception**
Gemini 2.0 Flash is used solely for visual extraction into strict Pydantic schemas.
- / Pure Python Math**
A mathematical engine (< 10ms) executes exact <=, >=, and tolerance delta math.
- / Verbatim Citation Gate**
Every violation is anchored to verbatim rulebook text. Unanchored flags are downgraded.
- / Sub-Second Latency**
Completes ingestion, hybrid retrieval, and full audit verification in < 0.35 seconds.

03 / SYSTEM BLUEPRINT

The 5-stage hybrid compliance pipeline

Decoupling probabilistic extraction from deterministic verification guarantees zero false positives with sub-second execution.

01

MULTI-MODAL INGESTION

PyMuPDF4LLM + RapidOCR ONNX

Ingests 10 to 30 page tech pack PDFs. Extracts layout tables, dimensions, bill of materials (BOM), and crops technical flats.

02

SCHEMA STRUCTURING

Gemini 2.0 Flash + Pydantic v2

Normalizes unstructured garment data into strongly-typed TechPackSpec models with strict unit conversions (cm, mm, GSM, FOB).

03

REGULATORY KNOWLEDGE BASE

Embedded Qdrant + BM25 Hybrid RAG

Dense semantic embeddings (FastEmbed 384-dim) combined with exact article BM25 keyword matching and Reciprocal Rank Fusion.

04

DETERMINISTIC VERIFIER GATE

Pure Python Verification Engine (< 10ms)

Calculates mathematical differences against FEI limits. Validates verbatim legal citations to eliminate false-positive flags.

05

SCORECARD & VENDOR ACTIONS

Next.js 14 Atelier Pro UI

Instant split-screen inspection with highlighted garment flats, delta reduction specs, and exportable A4 compliance certificates.

04 / CASE STUDY & PROOF

Auditing a 15-page tech pack in 350 milliseconds

Demonstrated on the Sovereign Grand Prix Competition Coat against official FEI Jumping and Dressage regulations.

0.35s

EXECUTION LATENCY

Sub-second turnaround SLA

54 / 54

AUTOMATED TESTS

Unit, integration & benchmarks

0.0%

FALSE POSITIVE RATE

Verbatim citation verified

AUDIT FINDINGS & SUPPLIER REMEDIATION

Collar Sponsor Logo Area

Observed: 65.0 cm² | Regulatory Ceiling: 60.0 cm²

Remediation: Downscale embroidery template by 7.7% before production run.

Over by +5.0 cm²

Lapel Contrast Velvet Piping

Observed: 3.5 mm | Regulatory Ceiling: 3.0 mm

Remediation: Substitute with 2.5 mm gold satin piping trim code TP-2026.

Over by +0.5 mm

Factory Quoted FOB Cost

Observed: \$62.00 | Regulatory Ceiling: \$55.00

Remediation: Commercial variance flag sent to Sourcing Lead for supplier negotiation.

Variance +\$7.00

05 / THE TECH STACK

Engineered for speed, privacy, and determinism

Built with modern Python and TypeScript without bloated external cloud dependencies.
Runs 100% locally or inside hardened Docker containers.

BACKEND API LAYER

FastAPI & Python 3.12

High-throughput async execution, strict Pydantic v2 schemas, automated OpenAPI documentation, and < 10ms execution.

FRONTEND INTERFACE

Next.js 14 App Router

Luxury split-screen desktop experience, synchronized technical flats, SVG inspection bounding boxes, and Quiet Luxury UI tokens.

EMBEDDED HYBRID RAG

Qdrant Vector DB

Local persistent on-disk vector store with HNSW indexing and BM25 exact keyword matching for zero cloud dependency.

MULTI-MODAL EXTRACTION

Gemini 2.0 Flash

Fast multimodal parsing for complex technical flat drawings, structured JSON generation, and aesthetic evaluation.

INGESTION & OCR

PyMuPDF4LLM + ONNX

Sub-second extraction of layout tables and vector sketches with RapidOCR fallback strictly for scanned pages.

TESTING & DEVOPS

Pytest & Docker CI

54 automated unit, integration, and benchmark tests with multi-stage non-root containerization.

06 / PRODUCTION BLUEPRINT

3 core lessons for production AI engineering

How to design AI applications when compliance failure is not an option.

01

LLMs for Perception, Code for Governance

Use neural networks where data is messy: PDF layouts, sketches, and natural language. Use deterministic code where logic is critical: legal thresholds, finances, and tolerances. Never let an LLM do arithmetic.

02

Verifier Gates Beat Prompt Engineering

Prompting an LLM to 'be accurate and strictly follow rules' is a brittle band-aid. Instead, structure model output with Pydantic and pass it through a hardcoded verification gate.

03

Ground Every Decision with Verbatim Citations

If a regulatory violation cannot be traced to exact verbatim text in the codified rulebook, downgrade it automatically to manual review. Eliminating false positives creates user trust.

Open-Source Implementation on GitHub:

github.com/KindlyGentleman/equestrian-compliance-auditor